JUSTINE NEEMA

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SUMMARY

Software Engineering student with hands-on experience designing full-stack applications, managing relational databases, and developing scalable backend systems using Python, Restful API, and FastAPI at African Leadership University. Skilled in technical research, system requirements analysis, and translating complex technical information and data into clear, structured documentation and strategic briefs. Experienced in evaluating software solutions, understanding system requirements, and documenting technical processes. Interested in applying software engineering, analytical research, and documentation skills to support digital transformation and improve public service delivery.

EDUCATIONAL BACKGROUND

Bsc (hons) Software Engineering **May 2024 – July 2027**
African Leadership University
Concentration: Full-stack development

PROFESSIONAL EXPERIENCE

Backend Engineer Intern July 2026 – Present
Rwanda ICT Chamber, Kigali

- Build and maintain backend systems and REST APIs in Python and PHP (Laravel) for internal and partner-facing applications.
- Design and maintain relational database schemas, supporting accurate and consistent stored data.
- Implementing backend business logic, validating incoming requests, and handling application errors to improve API reliability
- Collaborating with development teams to analyze technical requirements, troubleshoot issues, and deliver maintainable solutions.
- Developing API endpoints and integrating backend services to support application functionality and seamless data exchange.

Full-stack developer and IT Support Specialist (Part-time) **March 2025 – Present**
Raise Them Foundation, Kigali, Rwanda

- Build and maintain backend systems and REST APIs in Python and PHP (Laravel) for internal and partner-facing applications.
- Design and maintain relational database schemas, supporting accurate and consistent stored data.
- Implementing backend business logic, validating incoming requests, and handling application errors to improve API reliability
- Collaborating with development teams to analyze technical requirements, troubleshoot issues, and deliver maintainable solutions.
- Design and develop the Raise Them Foundation website.
- Design and develop the Raise Them Foundation e-Learning platform
- Digitise paper- and spreadsheet-based workflows into structured data systems to improve reporting accuracy.
- Manage operational databases and resolve data-quality issues at the source

Software Engineer (Backend) **January 2026 – June 2026**
Cylacon LTD, Kigali, Rwanda

- Developed backend systems in Python with PostgreSQL and MySQL for business applications.
- Designed RESTful APIs and defined data contracts between backend services and frontend clients.

- Improved database queries and schema design to support application performance.
- Tested, debugged,d and deployed changes, prioritising fixes by business impact.

Frontend Developer **September 2025 – December 2025**
Jackal Tech LTD, Kigali, Rwanda

- Built responsive web interfaces using JavaScript (ES6+), HTML5 and CSS3, integrating frontend features with backend APIs.
- Developed forms and interactive components to capture structured information and reduce downstream data-cleaning work.

PROJECTS

Raise Them Foundation E-learning platform **August 2026 – November 2026**
React with Typescript and Python/Django| PostgreSQL

- Developed an e-learning platform enabling learners to access courses, complete quizzes, track learning progress, and earn certificates.
- Implemented course management functionality to organize learning materials and make educational content accessible to learners.
- Developed quiz and assessment modules to evaluate learners' understanding and track their performance.
- Designed and managed database structures to support user accounts, course enrollment, assessments, learning progress, and certification records.
- Integrated certificate generation functionality to recognize learners who successfully completed course requirements.

Kanban Board Backend System **August 2026 – September 2026**
Python/Django| PostgreSQL

- Built backend services for a project-management tool with CRUD operations for boards, columns, tasks, and comments. • Implemented worker management features for assigning tasks, tracking ownership, and moving work through stages.
- Secured API endpoints with authentication and role-based access control, using a relational schema designed around the application's data needs.

Personnel Attendance Management System **August 2025 – October 2025**
Python/Django, JavaScript|SQL

- Designed the data model, including entities, relationships, and constraints, then developed backend services and the user interface.
- Handled attendance data edge cases such as duplicate entries, missing records,s and corrections.

SKILLS

- **Computer languages (advanced):** Java, PHP, Python, Linux.
- **Web application tools:** HTML, CSS, JavaScript.
- Mobile Application tools: Flutter, React native Expo Go
- Frontend Frameworks: ReactJs, NextJs
- Backend Frameworks: NodeJs with Express, NestJs, Laravel, Django
- APIs: Restful APIs, FastAPIs, GraphAPI
- Devops: CI/CD pipelines, GitHub actions, Git& GitHub version control and basic linux

- Caching: Redix, Load Balancers
- Database: PostgreSQL, MongoDB, MySQL, SQL

INTERESTS & ACHIEVEMENTS

- Exploring emerging software engineering practices and backend architecture through independent technical projects and practical experimentation.
- Studying digital transformation and e-government innovations, with particular interest in how technology improves public service delivery in Rwanda.
- Following developments in data analytics, database technologies, and API integration to broaden technical knowledge and understand real-world applications.
- Participating in continuous professional development through technical learning, software development exercises, and exploration of new development tools.